

Study of Activate function and IR pole

Aim:

To study different activation function and understand their role in deep learning module.

Objective:

1. To understand the importance of non-linearity introduced by activation function.
2. To compare their behaviour and significance in training deep learning model.
3. To evaluate how different activation function affect model performance.

Pseudocode:

Import required libraries:

Define mathematical function for sigmoid, Tanh, ReLU, Leaky ReLU, softmax

generate a range of input value (x)

compute outputs of all activation functions.

plot graph of each activation function.

Activation	Output range	Advantage	Use case
Sigmoid	$(0, 1)$	Smooth, probabilistic output	Binary classification
Tanh	$(-1, 1)$	Central amounts, better than Sigmoid	Hidden layer, older network.
Relu	$(0, \infty)$	Fast, reduce computation time	Hidden layer (modern NN / ANN)
Leaky Relu	$(-\infty, \infty)$	First Relu (devises issues)	Deep hidden layer.
Softmax	$(0, 1)$ Sum = 1	Given probabilistic distribution	Output layer for multi class.

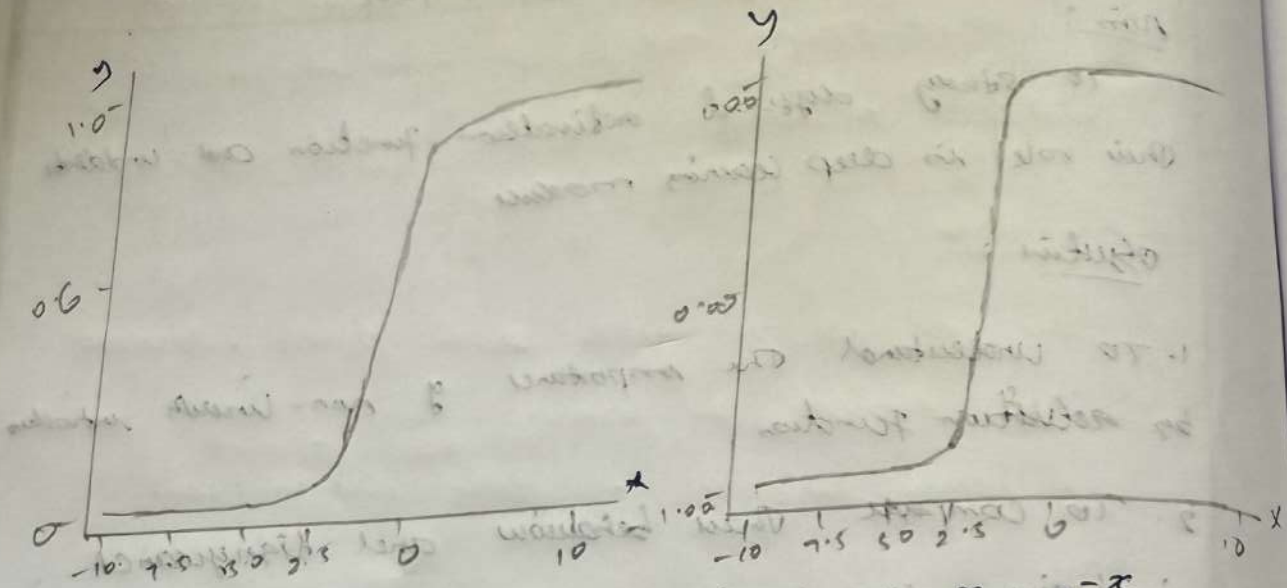
Observation:

- + Relu and Leaky Relu are most effective in hidden layer
- + Sigmoid and Tanh are used today due to vanishing gradient.
- + Softmax for multi class classification problem.

Result:

Studied different activation function & their use.

Sigmoid



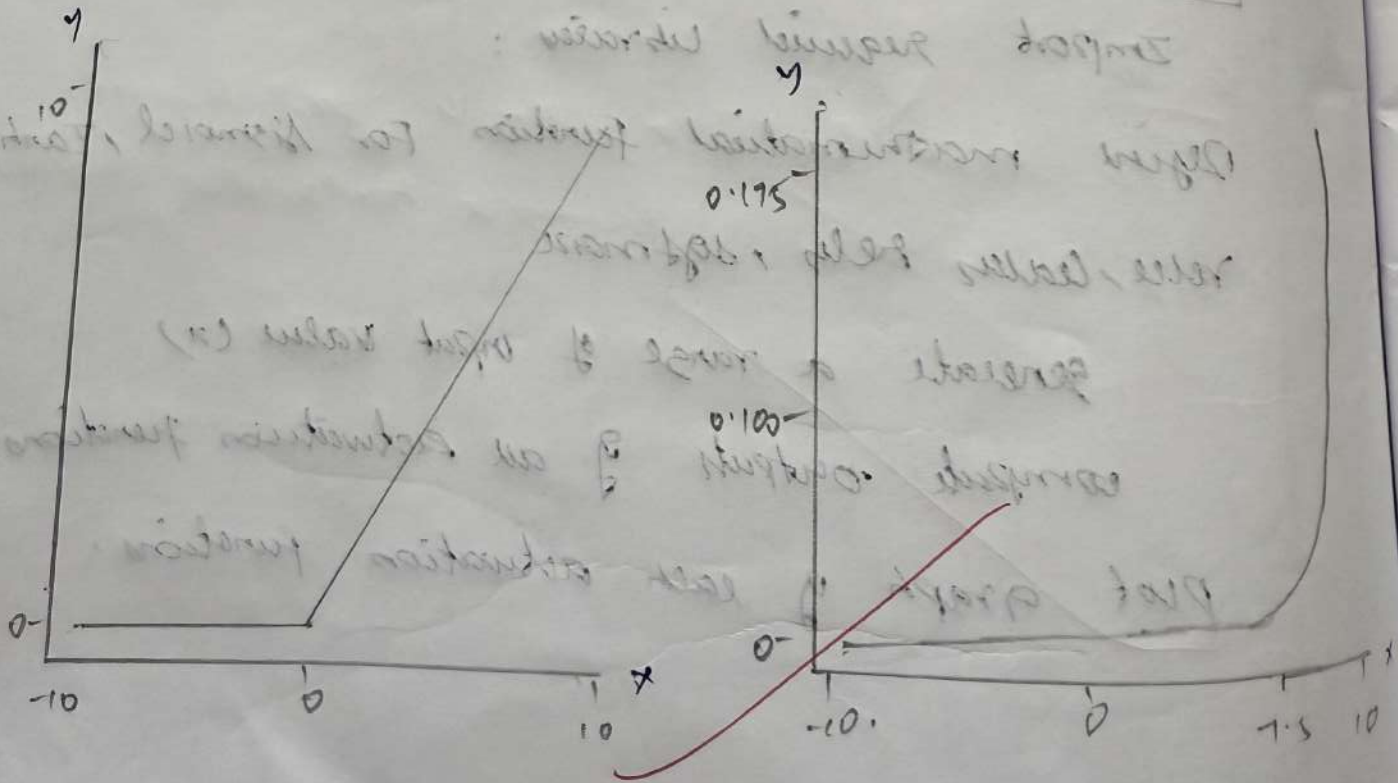
$$\frac{1}{1 + e^{-x}}$$

$$e^x - e^{-x}$$

$$e^x + e^{-x}$$

Relu

Softmax



$$\text{Relu}(x) = \max(0, x)$$

$$S_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

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lab5.ipynb

```
[4]: import torch
import torch.nn.functional as F

[3]: x = torch.tensor([-2.0, -1.0, 0.0, 1.0, 2.0])

[5]: sigmoid = torch.sigmoid(x)
tanh = torch.tanh(x)
relu = F.relu(x)
leaky_relu = F.leaky_relu(x, negative_slope=0.1)
softmax = F.softmax(x, dim=0)

[6]: print("Input:", x)
print("Sigmoid:", sigmoid)
print("Tanh:", tanh)
print("ReLU:", relu)
print("Leaky ReLU:", leaky_relu)
print("Softmax:", softmax)

Input: tensor([-2., -1., 0., 1., 2.])
Sigmoid: tensor([0.1192, 0.2689, 0.5000, 0.7311, 0.8808])
Tanh: tensor([-0.9640, -0.7616, 0.0000, 0.7616, 0.9640])
ReLU: tensor([0., 0., 0., 1., 2.])
Leaky ReLU: tensor([-0.2000, -0.1000, 0.0000, 1.0000, 2.0000])
Softmax: tensor([0.0117, 0.0317, 0.0861, 0.2341, 0.6364])
```

Simple 0 6 Python 3 (ipykernel) Idle Mem: 404.59 MB

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